

Appendix A

Exponential Families

Exponential families are the main connecting idea of classical statistics. Here we will provide a very brief sketch of the theory as it applies to the previous material, particularly that in Chapter 5. Advanced texts such as Lehmann and Romano (2005b) discuss exponential families, and their applications to testing and estimation, in more detail.

A one-parameter exponential family $\mathcal{F}$ is a family of probability densities on a subset $\mathcal{X}$ of the real line,

$$f_{\eta}(x) = e^{\eta x - \psi(\eta)} f_0(x) \quad (x \in \mathcal{X}) \quad (\text{A.1})$$

where η is the *natural* or *canonical* parameter, x is the *sufficient statistic*, $f_0(x)$ is the *carrier*, $\mathcal{X}$ is the *sample space*, and $\psi(\eta)$ is the *cumulant generating function*, or more simply the *normalizer*, so called because $\exp\{-\psi(\eta)\}$ is the positive constant required to make $f_{\eta}(x)$ integrate to 1,

$$e^{\psi(\eta)} = \int_{\mathcal{X}} e^{\eta x} f_0(x) dx. \quad (\text{A.2})$$

The family $\mathcal{F}$ consists of all exponential “tilts” of $f_0(x)$, $e^{\eta x} f_0(x)$, normalized to integrate to 1. Exponential tilting enjoys a crucial statistical property: if $x_1, x_2, \dots, x_N$ is an i.i.d. sample from $f_{\eta}(x)$, then $\bar{x} = \sum x_i / N$ is a sufficient statistic for estimating η , with densities

$$f_{\eta}^{\bar{x}}(\bar{x}) = e^{N[\eta \bar{x} - \psi(\eta)]} f_0^{\bar{x}}(\bar{x}), \quad (\text{A.3})$$

again a one-parameter exponential family. This greatly simplifies inference theory for η , the original motivation for definition (A.1) in the 1930s. Perhaps surprisingly, it then turned out that most familiar statistical families — normal binomial, Poisson, gamma — were in fact of form (A.1).

As a first example, $x \sim \mathcal{N}(\mu, 1)$ has density functions

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\mu)^2} = e^{\mu x - \mu^2/2} \cdot \varphi(x) \quad (\text{A.4})$$

for μ any value in $(-\infty, \infty)$, where $\varphi(x)$ is the standard normal density $\exp\{-x^2/2\}/\sqrt{2\pi}$. This is of form (A.1), having sufficient statistic x , natural parameter $\eta = \mu$, $\mathcal{X}$ equaling the whole real line, $\psi(\eta) = \eta^2/2$, and carrier $\varphi(x)$. (So the entire normal family (A.4) can be obtained from exponential tilts of $\varphi(x)$.) Repeated sampling $x_1, x_2, \dots, x_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$ as in (A.3) produces essentially the same family, but now with sufficient statistic $\bar{x}$.

The sample space for the Poisson family is the non-negative integers $\mathcal{X} = \{0, 1, 2, \dots\}$. Its (discrete) density function can be written in its usual form as

$$f(x) = e^{-\mu} \mu^x / x! = e^{(\log \mu)x - \mu} / x! \quad (\text{A.5})$$

for any $\mu > 0$. This is again of type (A.1), with

$$\eta = \log \mu, \quad \psi(\eta) = e^\eta, \quad (\text{A.6})$$

and $f_0(x) = 1/x!$. (It is only necessary for $f_0(x)$ to be positive, not to integrate to 1.)

The cumulant generating function is so named because differentiating it gives the cumulants of x . In particular, the mean and variance are the first two derivatives with respect to η , the usual notation being

$$\mu \equiv E_\eta\{x\} = \partial\psi(\eta)/\partial\eta \quad (\text{A.7})$$

and

$$V \equiv \text{var}_\eta\{x\} = \partial^2\psi(\eta)/\partial\eta^2. \quad (\text{A.8})$$

For the Poisson family, where $\psi(\eta) = \exp(\eta)$, we get

$$\mu = e^\eta \quad \text{and} \quad V = e^\eta, \quad (\text{A.9})$$

that is, the variance equals the expectation, and in fact all the higher cumulants equal μ . Note that formulas (A.7) and (A.8) do not involve $f_0(x)$ or any integration, which, if $\psi(\eta)$ is available, makes them wonderfully convenient for calculating moments.

The *deviance function* $D(\eta_1, \eta_2)$ is a measure of distance between two members of an exponential family,

$$D(\eta_1, \eta_2) = 2 \cdot E_{\eta_1} \left\{ \log \left(\frac{f_{\eta_1}(x)}{f_{\eta_2}(x)} \right) \right\} = 2 \int_{\mathcal{X}} \log \left(\frac{f_{\eta_1}(x)}{f_{\eta_2}(x)} \right) f_{\eta_1}(x) dx. \quad (\text{A.10})$$

It is always non-negative, but usually not symmetric, $D(\eta_1, \eta_2) \neq D(\eta_2, \eta_1)$. $(D(\eta_1, \eta_2))/2$ is also known as the *Kullback–Leibler distance*, another related term being *mutual information*.) Deviance generalizes the concept of

squared error distance $(\eta_1 - \eta_2)^2$ or $(\mu_1 - \mu_2)^2$, this being $D(\eta_1, \eta_2)$ in the normal family (A.4). For the Poisson family,

$$D(\eta_1, \eta_2) = 2\mu_1 \left[\log \left(\frac{\mu_1}{\mu_2} \right) - \left(1 - \frac{\mu_2}{\mu_1} \right) \right]. \quad (\text{A.11})$$

In a regression situation such as (5.15)–(5.17), minimizing the sum of deviances between the observations and a parameterized regression function is equivalent to maximum likelihood estimation. This amounts to least squares fitting in a normal regression problem, but “least deviance” is more efficient in a Poisson regression context like (5.17).

A.1 Multiparameter Exponential Families

By changing η and x in (A.1) to K -dimensional vectors $\boldsymbol{\eta}$ and $\boldsymbol{x}$ we obtain a K -parameter exponential family of densities

$$f_{\boldsymbol{\eta}}(\boldsymbol{x}) = e^{\boldsymbol{\eta}'\boldsymbol{x} - \psi(\boldsymbol{\eta})} f_0(\boldsymbol{x}). \quad (\text{A.12})$$

Now the factor $\exp\{\sum_1^K \eta_k x_k\}$ can tilt the carrier $f_0(\boldsymbol{x})$ in K possible directions. The sufficient statistic $\boldsymbol{x}$ is usually some K -dimensional function of a more complicated set of data. Formula (A.12) provides a convenient way for the statistician to model complex data sets flexibly, by focusing on K summary statistics x_k , for example powers of z in (5.10).

Once again, most of the familiar multiparameter families — normal, Dirichlet, multinomial — are of form (A.12). The multinomial family was seen to play an important role in Section 5.2, where count y_k was the number of the N z -values falling into the k th bin, for $k = 1, 2, \dots, K$. If the z_i are independent, vector $\boldsymbol{y} = (y_1, y_2, \dots, y_K)'$ follows a multinomial distribution on K categories, sample size N ,

$$\boldsymbol{y} \sim \text{mult}_K(N, \boldsymbol{\pi}) \quad (\text{A.13})$$

with

$$\boldsymbol{\pi} = (\pi_1, \pi_2, \dots, \pi_K)' \quad (\text{A.14})$$

the vector of bin probabilities, $\pi_k = \Pr\{z \in \mathcal{Z}_k\}$ in notation (5.11).

The multinomial density function

$$f(\boldsymbol{y}) = c(\boldsymbol{y}) \prod_{k=1}^K \pi_k^{y_k} \quad \left[c(\boldsymbol{y}) = N! / \prod_1^K y_k! \right] \quad (\text{A.15})$$

is a $(K - 1)$ -dimensional exponential family, one degree of freedom being

lost to the constraint $\sum_1^K \pi_k = 1$. A convenient parameterization lets $\boldsymbol{\eta} = (\eta_1, \eta_2, \dots, \eta_K)'$ be any vector in K -dimensional Euclidean space $\mathcal{R}^K$, and defines

$$\pi_k = e^{\eta_k} / \sum_1^K e^{\eta_j} \quad \text{for } k = 1, 2, \dots, K. \quad (\text{A.16})$$

Then $\log(\pi_k) = \eta_k - \log(\sum_1^K e^{\eta_j})$, and (A.15) takes the form of (A.12),

$$f_{\boldsymbol{\eta}}(\mathbf{y}) = e^{\boldsymbol{\eta}'\mathbf{y} - \psi(\boldsymbol{\eta})} c(\mathbf{y}) \quad \left[\psi(\boldsymbol{\eta}) = \log \sum_1^K e^{\eta_j} \right]. \quad (\text{A.17})$$

This looks like a K -parameter family, but every scalar multiple $m \cdot \boldsymbol{\eta}$ produces the same vector $\boldsymbol{\pi}$ in (A.16) and (A.13), removing one degree of freedom.

Moment properties (A.7)–(A.8) extend directly to multivariate exponential families,

$$\boldsymbol{\mu} = E_{\boldsymbol{\eta}}\{\mathbf{x}\} = (\dots \partial \psi / \partial \eta_k \dots)' \quad (\text{A.18})$$

and

$$V = \text{cov}_{\boldsymbol{\eta}}\{\mathbf{x}\} = (\dots \partial^2 \psi / \partial \eta_j \partial \eta_k \dots), \quad (\text{A.19})$$

the last notation indicating a $K \times K$ matrix of second derivatives. As in (A.10), the deviance function is

$$D(\boldsymbol{\eta}_1, \boldsymbol{\eta}_2) = 2E_{\boldsymbol{\eta}_1} \{ \log(f_{\boldsymbol{\eta}_1}(\mathbf{x}) / f_{\boldsymbol{\eta}_2}(\mathbf{x})) \}. \quad (\text{A.20})$$

Exponential families and maximum likelihood estimation are closely related, both historically and mathematically. The “score function” equals

$$\frac{\partial \log f_{\boldsymbol{\eta}}(\mathbf{x})}{\partial \boldsymbol{\eta}} = \left(\dots \frac{\partial \log f_{\boldsymbol{\eta}}(\mathbf{x})}{\partial \eta_k} \dots \right)' = \mathbf{x} - \boldsymbol{\mu} \quad (\text{A.21})$$

from (A.12) and (A.18), so the MLE $\hat{\boldsymbol{\eta}}$ is the value of $\boldsymbol{\eta}$ making the expectation vector $\boldsymbol{\mu}$ equal the observed vector $\mathbf{x}$. The second derivative matrix is

$$\frac{\partial^2 \log f_{\boldsymbol{\eta}}(\mathbf{x})}{\partial \boldsymbol{\eta}^2} = \left(\dots \frac{\partial^2 \log f_{\boldsymbol{\eta}}(\mathbf{x})}{\partial \eta_j \partial \eta_k} \dots \right) = -V \quad (\text{A.22})$$

according to (A.18)–(A.19). This implies that $\log f_{\boldsymbol{\eta}}(\mathbf{x})$ is a concave function of $\boldsymbol{\eta}$ (since the covariance matrix V is positive semidefinite for every choice of $\boldsymbol{\eta}$), usually making it straightforward to calculate iteratively the MLE $\hat{\boldsymbol{\eta}}$.

These days multivariate exponential families play their biggest role in

generalized linear models. A GLM supposes that the K -vector of natural parameters $\boldsymbol{\eta}$ in (A.12) is a known linear function of a J -vector $\boldsymbol{\beta}$ ($J \leq K$), say

$$\boldsymbol{\eta} = \underset{K \times J}{M} \boldsymbol{\beta}. \quad (\text{A.23})$$

This construction extends the notion of ordinary linear models to non-normal exponential families. It reduces the K -parameter family (A.12) to a J -parameter exponential family, with natural parameter vector $\boldsymbol{\beta}$ and sufficient statistic $M' \mathbf{x}$. The MLE equations for $\boldsymbol{\beta}$ take the form

$$M'(\mathbf{x} - \boldsymbol{\mu}(\boldsymbol{\beta})) = \mathbf{0} \quad (\text{A.24})$$

where $\boldsymbol{\mu}(\boldsymbol{\beta})$ is the expectation vector in (A.12) corresponding to $\boldsymbol{\eta} = M\boldsymbol{\beta}$. McCullagh and Nelder (1989) is the standard reference for generalized linear models.

A.2 Lindsey's Method

Model (5.10) is a J -parameter exponential family with natural parameter $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_J)'$, sufficient statistic $\mathbf{x} = (z, z^2, \dots, z^J)'$, and normalizer $\psi(\boldsymbol{\beta})$ equaling $-\beta_0$. If $z_1, z_2, \dots, z_N$ is an i.i.d. sample from $f(z)$ then, as in (A.3), we still have a J -parameter exponential family, now with sufficient statistic $\bar{\mathbf{x}} = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_J)'$,

$$\bar{x}_j = \sum_{i=1}^N z_i^j / N, \quad (\text{A.25})$$

and natural parameter vector $N\boldsymbol{\beta}$.

Solving for the MLE $\hat{\boldsymbol{\beta}}$ requires special programming, essentially because there is no closed form for $\psi(\boldsymbol{\beta})$. Lindsey's method (Lindsey, 1974; Efron and Tibshirani, 1996, Sect. 2) is a way of using standard Poisson GLM software to find $\hat{\boldsymbol{\beta}}$. The N z -values are binned into K counts $\mathbf{y} = (y_1, y_2, \dots, y_K)'$ as in (5.11)–(5.12) so that the density $f_{\boldsymbol{\beta}}(z)$ is reduced to a multinomial probability vector $\boldsymbol{\pi}(\boldsymbol{\beta})$ (A.14), with $\pi_k(\boldsymbol{\beta}) = d \cdot f_{\boldsymbol{\beta}}(x_k)$ in notation (5.14).

A special relationship between multinomial and Poisson distributions is then invoked: assuming that the y_k 's are independent Poisson observations,

$$y_k \stackrel{\text{ind}}{\sim} \text{Poi}(\nu_k), \quad \nu_k = N\pi_k(\boldsymbol{\beta}) \quad \text{for } k = 1, 2, \dots, K, \quad (\text{A.26})$$

the MLE $\hat{\boldsymbol{\beta}}$ in (A.26) equals that in the multinomial model of (A.13),

$\mathbf{y} \sim \text{mult}_k(N, \boldsymbol{\pi}(\boldsymbol{\beta}))$. Family (5.10), for example, becomes a Poisson GLM (A.23) in this formulation, with $\eta_k = \log(\nu_k)$ and M having k th row

$$(x_k, x_k^2, \dots, x_k^J)', \quad (\text{A.27})$$

x_k being the k th bin midpoint (5.13).

In computing language R, the single call

$$\text{glm}(\mathbf{y} \sim M, \text{family}=\text{Poisson}) \quad (\text{A.28})$$

produces $\hat{f}$, the MLE for density $f(z)$ in (5.10), evaluated at $z = x_k$ for $k = 1, 2, \dots, K$. These are the heavy curves plotted, for example, in Figure 5.1 and Figure 6.1. Calling $\hat{f}$ the MLE assumes that the z_i are independent, which is usually untrue; $\hat{f}$ is better thought of as a smooth parametric fit to the histogram heights y_k , minimizing the total Poisson deviance (5.17) as a fitting criterion.